EXPLAINABLE ARTIFICIAL INTELLIGENCE AS A BRIDGE BETWEEN BLACK-BOX MODELS AND SCIENTIFIC TRANSPARENCY IN ATMOSPHERIC AND SPACE SCIENCES

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ABSTRACT

Artificial Intelligence (AI) is increasingly transforming atmospheric and space sciences by enabling the analysis of complex datasets and improving predictions in climate modelling, weather forecasting, satellite monitoring, spacecraft anomaly detection, and astrophysical research. However, deep learning models often function as "black boxes," limiting transparency and reducing confidence in their predictions. This survey, based on peer-reviewed journal articles, conference papers, and authoritative reviews published between 2020 and 2026, examines the role of Explainable Artificial Intelligence (XAI) in addressing these challenges. Techniques including feature importance, sensitivity analysis, visualization, counterfactual and prototype-based explanations, rule-based models, and transparent documentation can improve interpretability, reliability, causality, fairness, privacy, and trust. By connecting data-driven AI with established physical principles, XAI strengthens scientific credibility, supports safer operational decisions, and promotes responsible AI adoption. The study argues that XAI is essential for advancing scientific discovery and societal resilience in atmospheric and space applications.

KEYWORDS

Explainable AI, Atmospheric Research, Space Sciences, Interpretability, Climate Modelling, Satellite Systems

1. INTRODUCTION

Artificial Intelligence (AI) has become increasingly vital in processing vast and diverse datasets ranging from satellite observations and sensor networks to geospatial records and climate models, to support critical decisions in atmospheric research and space sciences. However, as these AI systems become more sophisticated, their decision-making processes often appear as "black boxes," making it difficult for scientists, policymakers, and stakeholders to interpret or trust their outputs [1,2]. Explainable Artificial Intelligence (XAI) emerges as a solution, providing methods that enhance transparency, interpretability, and trust in AI-driven predictions and analyses. In these scientific domains, where accuracy and accountability are paramount, XAI ensures that model outputs are comprehensible and analyzable without requiring deep expertise in AI. Understanding how models arrive at specific forecasts, anomaly detections, or space-event

predictions is essential for building trust and acceptance among researchers, governments, and society at large. By making AI systems interpretable, XAI also addresses concerns of bias, fairness, and unintended errors, enabling scientists and regulators to evaluate ethical implications, validate findings, and strengthen resilience in decision-making.

Explainable Artificial Intelligence (XAI) offers a pathway to bridge the gap between the complexity of advanced AI models and the need for transparency, trust, and scientific rigor in atmospheric and space sciences. As AI becomes increasingly central to tasks such as weather prediction, climate modeling, satellite data analysis, and space mission planning, the ability to interpret and explain its outputs is essential. Enhancing interpretability not only strengthens accountability and ethical use but also enables scientists, policymakers, and stakeholders to critically evaluate AI-driven insights, fostering a more responsible and trustworthy integration of AI into research and operational systems [3]. AI models in these fields often function as "black boxes," especially when applied to high-dimensional climate datasets or complex remote sensing imagery, where it is challenging to explain how predictions, such as extreme weather conditions, atmospheric anomalies, or spacecraft navigation risks, are generated. Cracking open this black box is vital for building trust in forecasts and analyses, ensuring that AI supports reliable decision-making in contexts where errors could have far-reaching consequences. The challenges of complex space data and stringent reliability and safety demands highlight the need for robust, interpretable AI systems, while discussing XAI techniques such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations), intrinsic approaches (such as decision trees), and generalized additive models [4].

Furthermore, transparency in training data, bias mitigation, and continuous validation are critical for ensuring trustworthy outcomes. Collaborative efforts among AI developers, atmospheric scientists, and space researchers play a key role in aligning AI systems with domain knowledge and human expectations. In the work of [5], emerging applications already demonstrate the potential of XAI in improving the interpretability of storm intensity forecasts, explaining satellite-based climate variability analyses, and clarifying anomaly detection in space missions which underscore the importance of XAI in strengthening transparency, accountability, and user confidence, ensuring that AI-driven tools in atmospheric and space sciences are not only powerful but also explainable and scientifically reliable. Despite the promising potential, [6] reported that XAI faces several challenges in enhancing the interpretability and understanding of AI models for humans, such as the inherent trade-off between model complexity and interpretability, as highly sophisticated models often sacrifice transparency for performance. Meanwhile, developing commonly accepted standards for interpretability poses a significant challenge, as different domains may require tailored explanations. Resolving the issue of context-aware explanations is another obstacle, as the interpretability of AI models should account for the specific needs and knowledge levels of diverse user groups. According to [7], striking a balance between accuracy and simplicity in explanation generation remains a persistent challenge, especially in complex, high-dimensional datasets. Furthermore, [8] reviewed that ensuring that XAI methods effectively handle real-world scenarios, including unforeseen situations and adversarial attacks, is essential for their practical applicability, and overcoming these challenges is imperative for the widespread adoption of XAI, ensuring that AI systems are not only accurate but also transparent, trustworthy, and accessible to a broad range of users.

Physics-Informed Neural Networks (PINNs) incorporate governing physical equations, conservation laws, and boundary conditions directly into the training process, ensuring that model predictions remain consistent with established atmospheric and space science principles rather than relying solely on data-driven correlations. Consequently, XAI applied to PINNs produces explanations that are more scientifically meaningful and less likely to highlight spurious

or non-physical relationships, thereby improving the transparency, reliability, and trustworthiness of AI models in atmospheric and space science applications. The studies in [42] highlight emerging approaches, including physics-informed models and enhanced uncertainty quantification techniques, as potential solutions for addressing ongoing challenges related to physical consistency and trustworthiness in machine learning applications.

This paper carried out research survey in explainable artificial intelligence (XAI), particularly the urgent need for models in atmospheric research and space sciences to be transparent and interpretable. The contributions of this study to the existing body of knowledge are as follows:

- To present a comprehensive summary of the surveyed reviewed literature on recent and emerging XAI techniques that enhance explanatory capacity in atmospheric and space science applications.
- To examine the explanatory depth of Explainable Artificial Intelligence (XAI) by evaluating how techniques such as feature importance, visualization, and counterfactual explanations enhance the clarity, interpretability, and trustworthiness of AI decision-making.
- To investigate and evaluate the effectiveness of specific XAI methods such as sensitivity analysis, rule-based models, documentation and transparency frameworks, prototype-based explanations, and visualization techniques in improving the interpretability of atmospheric and space science models.
- To identify and analyze the key challenges associated with implementing XAI techniques in these scientific domains, including data complexity, uncertainty quantification, scalability, and the integration of domain expertise.

The remaining part of this paper is organized into the following sections: Section 2 covers the methodology used for the survey and the criteria for the review of related literature which was summarized in a table. This is followed by the explanatory depth of XAI in section 3 while section 4 outlines techniques and methods to enhance the interpretability and understandability of AI models in atmospheric and space science, and section 5 deals with some challenges and solutions of implementing XAI in atmospheric and space sciences, such as data complexity, uncertainty quantification, scalability, and integration of domain expertise. Finally, Section 6 is the conclusion and some areas of open research direction related to XAI in the sciences.

2. RESEARCH METHODOLOGY FOR THE SURVEY

The research survey was based on selected literature which indicates that XAI is increasingly being adopted to improve the interpretability, transparency, trustworthiness, and scientific credibility of AI models. While earlier studies focused on generic XAI frameworks and methods, more recent publications (2024–2026) emphasize domain-specific applications in atmospheric science, climate prediction, environmental monitoring, hydrology, and space science, demonstrating that explainable AI serves as a critical bridge between high-performing black-box models and scientifically transparent decision-making. This trend supports the proposed review topic by highlighting the growing role of XAI in making AI-driven atmospheric and space science research more interpretable, reproducible, and trustworthy. This is summarized and presented in Table 1.

The selection of studies for this survey was guided by predefined inclusion and exclusion criteria to ensure methodological rigor, relevance, and reliability. Only peer-reviewed journal articles, conference papers, and authoritative review papers published between 2020 and 2026 were considered. This period was selected because explainable artificial intelligence (XAI) has

experienced rapid methodological development during these years, particularly regarding transparency, interpretability, evaluation metrics, and applications in scientific domains. Recent literature provides more comprehensive taxonomies, evaluation frameworks, and practical implementations than earlier studies, making it appropriate for capturing the current state of the field.

Table 1. Summary of the Surveyed Literature

Authors	Year	Title of Publication	Relevance to the Topic
Taghikhah et al.	2022	Machine-assisted agent-based modeling: Opening the black box	Demonstrates methods for improving transparency in black-box models through interpretable agent-based modeling.
Bommer et al.	2024	Finding the Right XAI Method—A Guide for the Evaluation and Ranking of Explainable AI Methods in Climate Science	Directly addresses XAI evaluation for climate science applications and provides guidance for selecting appropriate XAI methods.
Kaushik	2023	Explainability in Machine Learning: Bridging the Gap Between Model Complexity and Interpretability	Explains how XAI balances predictive performance with interpretability, providing theoretical foundations.
Agarwal et al.	2025	Developing Explainable Artificial Intelligence Models for Space Science Applications	Demonstrates XAI techniques specifically designed for space science decision-making and scientific interpretation.
Schiller et al.	2025	Artificial Intelligence in Environmental and Earth System Sciences: Explainability and Trustworthiness	Discusses explainability and trustworthy AI for Earth and environmental sciences, highlighting transparency challenges.
Saeed and Omlin	2023	Explainable AI (XAI): A Systematic Meta-Survey of Current Challenges and Future Opportunities	Comprehensive survey summarizing XAI methods, challenges, and research directions.
Chamola et al.	2023	A Review of Trustworthy and Explainable Artificial Intelligence (XAI)	Reviews trustworthy AI principles including explainability, fairness, and transparency relevant to scientific applications.
Chromik et al.	2021	I Think I Get Your Point, AI! The Illusion of Explanatory Depth in Explainable AI	Explores how users perceive AI explanations and limitations of explanation effectiveness.
Hashemi	2023	Who Wants What and How: A Mapping Function for Explainable Artificial Intelligence	Presents stakeholder-oriented XAI frameworks for different user needs.
Kalasampath et al.	2025	A Literature Review on Applications of Explainable Artificial Intelligence (XAI)	Summarizes XAI applications across domains, providing a broad foundation for atmospheric and space science studies.
Antoniadi et al.	2021	Current Challenges and Future Opportunities for XAI in Machine Learning-Based Clinical Decision Support Systems	Highlights common XAI challenges transferable to scientific decision-support systems.
Morrison et al.	2023	Evaluating the Impact of Human Explanation Strategies on Human-AI Visual Decision-Making	Investigates how explanations improve human understanding and trust in AI predictions.
Sobhana et al.	2025	XAI-Based Heavy Rainfall Prediction Using Satellite Data in Andhra Pradesh	Demonstrates practical use of XAI for satellite-based atmospheric prediction.

Zacharias et al.	2022	Designing a Feature Selection Method Based on Explainable Artificial Intelligence	Shows how XAI improves feature selection and model interpretability.
Minh et al.	2022	Explainable Artificial Intelligence: A Comprehensive Review	Comprehensive review of XAI techniques, taxonomy, and applications.
Stepin et al.	2021	A Survey of Contrastive and Counterfactual Explanation Generation Methods for Explainable Artificial Intelligence	Reviews advanced explanation techniques applicable to scientific AI models.
Cortez and Embrechts	2023	Using Sensitivity Analysis and Visualization Techniques to Open Black Box Data Mining Models	Introduces visualization and sensitivity analysis methods to interpret black-box models.
Vilone et al.	2020	A Comparative Analysis of Rule-Based, Model-Agnostic Methods for Explainable Artificial Intelligence	Compares model-agnostic explanation methods suitable for scientific applications.
Kale et al.	2023	Provenance Documentation to Enable Explainable and Trustworthy AI	Discusses provenance and reproducibility, supporting scientific transparency.
Bobek et al.	2022	Enhancing Cluster Analysis with Explainable AI and Multidimensional Cluster Prototypes	Demonstrates explainable clustering techniques for data analysis.
Alicioglu and Sun	2022	A Survey of Visual Analytics for Explainable Artificial Intelligence Methods	Reviews visualization techniques that improve interpretation of AI decisions.
Mathew et al.	2025	Recent Emerging Techniques in Explainable Artificial Intelligence to Enhance the Interpretability and Understanding of AI Models for Human	Reviews recent advances in interpretable AI techniques applicable across scientific domains.
Gholami et al.	2024	An Interpretable Model Based on Machine Learning and SHAP Interpretation Technique for Mapping Wind Erosion Hazard	Applies SHAP explanations to environmental hazard mapping, demonstrating atmospheric relevance.
Patel and Shah	2024	Feature Importance in Machine Learning with Explainable Artificial Intelligence (XAI) for Rainfall Prediction	Demonstrates feature importance analysis for rainfall forecasting.
Hao et al.	2024	Insight into Glacio-Hydrological Processes Using Explainable Machine-Learning (XAI) Models	Uses XAI to explain hydrological processes, supporting Earth system science transparency.
Gómez-Pérez et al.	2022	Artificial Intelligence and Natural Language Processing and Understanding in Space: A Methodological Framework and Four ESA Case Studies	Discusses AI methodologies for space science with emphasis on explainability and understanding.
Al Mazroa et al.	2024	Computer Vision with Explainable Artificial Intelligence for Visual Pollution Detection in the Kingdom of Saudi Arabia	Demonstrates interpretable computer vision techniques using XAI.
Zaidan et al.	2025	Artificial Intelligence for Atmospheric Sciences: A Research Roadmap	Provides future directions for AI in atmospheric sciences, including explainability needs.
Materia et al.	2024	Artificial Intelligence for Climate Prediction of Extremes: State of the Art, Challenges, and Future Perspectives	Reviews AI applications in climate prediction and discusses transparency challenges.
Allen	2025	Explainable AI (XAI) and Autonomous Systems for Persistent Human-Machine Operations in Space	Explores XAI for autonomous space missions requiring trustworthy decision-making.
Silva and	2024	Limitations of XAI Methods for Process-	Critically evaluates the limitations of

Keller		Level Understanding in the Atmospheric Sciences	current XAI techniques in atmospheric science.
Jahan et al.	2025	Uncertainty Quantification of Wind Gust Predictions Using Evidential Neural Networks and Explainable AI	Combines uncertainty quantification and XAI for atmospheric prediction reliability.
Desai	2024	Explainable AI for Urban Air Quality Forecasting: Integrating Satellite, Traffic, and IoT Sensor Data	Demonstrates XAI for environmental monitoring and urban atmospheric forecasting.
Hrast Essenfelder et al.	2025	Expert-Driven Explainable Artificial Intelligence Models Can Detect Multiple Climate Hazards Relevant for Agriculture	Illustrates expert-guided XAI for climate hazard detection and interpretable environmental decision-making.

3. EXPLANATORY DEPTH OF EXPLAINABLE AI

In the context of XAI, explanatory depth refers to the extent of detail and clarity in the explanations offered by AI systems. Some works have provided explanatory depth on XAI; [9] stated that explanatory depth shows how deeply an AI system can explain its decisions, going beyond seeming or shallow explanations. The concept of explanatory depth is vital because not all AI explanations are equally informative or useful. Some AI systems may provide high-level, imprecise explanations that do not help users understand the underlying reasoning, while others can offer detailed, insightful explanations that enhance users' trust and comprehension [10]. So, [11] stated that XAI gives the richness and detail of the explanations provided by AI systems in making AI more understandable and trustworthy, and achieving it involves various techniques and considerations to ensure that the explanations are meaningful and helpful to users. There are various methods used to achieve explanatory depth in XAI, and some of these techniques include: feature importance, visualization, and counterfactual explanations

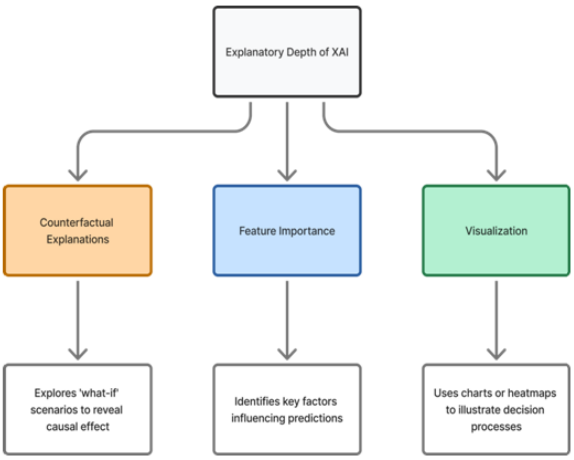


Figure 1. Explanatory Depth of Explainable AI (XAI)

From Figure 1, feature importance identifies major factors and key variables that influence predictions. According to [12], feature importance is the most influential factor in the decision-making process, whereas [13] demonstrated that visualization techniques employing visual aids effectively explain the decision-making process and the interactions among features and [14] explained that counterfactual explanations is about what changes in input data would have resulted in a different decision and local verse global explanations that offer insight into how a specific decision was made (local) as well as how the overall model behaves (global). These XAI techniques enhance explanatory depth by enabling users to comprehend and trust the decisions

made by AI systems in atmospheric and space sciences. According to [15], the feature importance technique seeks to improve explanatory depth by pinpointing the most significant features influencing an AI model's decisions. By determining which features most affect model predictions, users gain insights into the factors driving the AI system's behaviour. Furthermore, it allows users to prioritize their attention and understand the underlying logic of the model so that users can focus on relevant factors and comprehend why certain decisions are made, thereby increasing the transparency and interpretability of the AI system. Specifically, feature importance is used to perform the following tasks: highlight which input variables most influence model predictions, enhance transparency and guide scientific interpretation, and identify dominant features like humidity, pressure, or solar wind density in forecasting models.

Secondly, visualization techniques play a crucial role in achieving explanatory depth by providing intuitive representations of AI model behaviour, by transforming complex model outputs into visually interpretable formats, such as charts, graphs, or heatmaps, users can better understand the patterns and relationships captured by the AI system [16]. Visualization techniques enable users to explore the inner workings of the model, identify trends, and detect atmospheric anomalies more effectively. Additionally, interactive visualizations allow users to manipulate data inputs and observe real-time changes in model predictions, facilitating a deeper understanding of how the AI system responds to different scenarios. While the work of [17] on the counterfactual explanations offers another avenue for achieving explanatory depth in XAI by providing insights into "what-if" scenarios by generating hypothetical instances where model predictions change, users can gain a better understanding of the decision boundaries and causal relationships within the AI system, which help users explore alternative outcomes and understand how changes in input variables influence model predictions. Counterfactual explanations also reveal causal boundaries and help understand why a model made a specific decision, as well as strengthen causal reasoning and interpretability. These three explanatory depth techniques assist in achieving the goals of atmospheric and space science Explainable AI as presented in Figure 1.

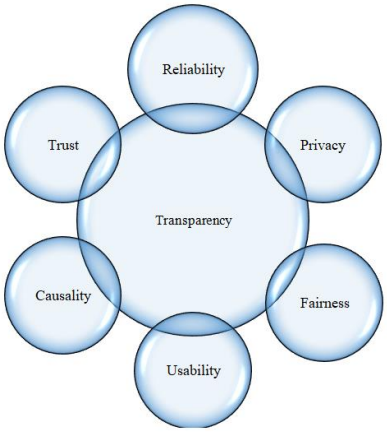


Figure 2. Goals of Explainable AI

Figure 2 illustrates that the primary objective of Explainable Artificial Intelligence (XAI) is to improve the comprehension and acceptance of AI systems. This is achieved by integrating essential principles such as transparency, reliability, causality, usability, privacy, trust, and fairness. Transparency ensures that the internal mechanisms of AI models are clear and understandable, promoting trust and accountability. Reliability focuses on consistently producing accurate and dependable results, contributing to the overall credibility of AI systems. Causality aims to uncover the cause-and-effect relationships within AI decisions, providing deeper insights

into model behaviour. Usability emphasizes making AI systems accessible and user-friendly, catering to diverse stakeholders.

Privacy concerns involve safeguarding sensitive information and ensuring responsible data handling practices. Trust is built through transparent processes, reliable outcomes, and ethical considerations. Fairness addresses the need for unbiased and equitable AI decisions, mitigating the impact of algorithmic biases, and thereby promoting ethical and responsible AI practices across various domains. Hence, the integration of these essential principles ensures that AI systems are understandable, accountable, and ethically aligned, thereby fostering confidence and responsible decision-making in scientific and operational contexts which would be the basis for the techniques and methods to make AI models more interpretable and understandable.

4. TECHNIQUES AND METHODS TO MAKE AI MODELS MORE INTERPRETABLE AND UNDERSTANDABLE

The importance of making AI models more interpretable and understandable in atmospheric and space science for building trust and ensuring the responsible deployment of AI systems is key. AI practitioners can employ certain techniques and methods to enhance model interpretability and understandability, fostering responsible AI deployment and increasing user trust. Table 2 presents some of these techniques and methods.

Table 2: Techniques, Description, Strengths, and Weaknesses for Making AI Interpretable and Understandable in Atmospheric and Space Sciences

S/N	Techniques/ Methods	Description	Strength	Weakness
1.	Sensitivity Analysis (SA)	Sensitivity analysis evaluates how changes in input variables affect the output of an AI model. By systematically varying parameters, it highlights which factors exert the most influence on predictions.	SA identifies critical atmospheric or space variables driving model predictions (e.g., humidity in rainfall forecasting, or solar wind density in satellite anomaly detection). Improves model transparency by showing direct cause and effect relationships. Supports uncertainty quantification in climate and space simulations, helping to distinguish robust predictions from noise.	Computationally intensive for high-dimensional data, typical in climate or space models. May oversimplify interactions when variables are highly correlated (e.g., temperature and pressure). Requires domain expertise to interpret results meaningfully; otherwise, insights may be misapplied [18].
2.	Rule-Based Models (RBM)	RBM is a technique in which the decision-making process of an AI system is explicitly represented as a set of rules. These rules take the form of "if-then" statements, actions, or predictions. It aligns with human logic,	RBM is easy to interpret, as scientists and mission operators can directly trace AI reasoning. Aligns well with established physical laws and domain heuristics (e.g., “if relative humidity > 90% and surface temperature drops, then	Limited scalability for complex, nonlinear systems (e.g., chaotic climate changes). May oversimplify nuanced atmospheric or orbital interactions. Performance often lags behind advanced deep learning models in

		making the model's decision process more interpretable and understandable.	precipitation is likely”). Useful for hybrid approaches, where AI augments physical-based models with empirical rules.	accuracy [19].
3.	Documentation and Transparency (D & T)	This involves structured documentation of AI models, including data sources, preprocessing, assumptions, limitations, and decision-making logic. Examples include “model cards” and “datasheets for datasets.”	Enhances reproducibility in climate and space science, where transparency of datasets is crucial. Helps policymakers and non-experts understand model limitations and appropriate use cases. Builds trust by openly disclosing biases, uncertainties, and operational constraints.	Relies on rigorous and consistent documentation practices, which can be resource-intensive. May still fail to capture indirect model behaviours or hidden biases. Less effective if users lack the technical expertise to interpret documentation [20].
4.	Prototype-Based Explanations (PBE)	Prototype-based methods explain AI decisions by comparing predictions with representative examples (prototypes) or contrasting cases (criticisms).	Intuitive for scientists, predictions can be linked to similar historical cases. Useful in satellite image analysis by linking detected anomalies to visually similar past observations. Supports training and validation by grounding AI predictions in known phenomena [21].	Quality depends on the representativeness of the training dataset; biased or incomplete data can mislead explanations. May oversimplify complex atmospheric or orbital dynamics by focusing too narrowly on case similarity. Can struggle when novel, previously unseen patterns occur in space weather events.
5.	Visualization	Enhances the interpretability and understanding of AI models by presenting their complex decision-making processes visually and intuitively. This method entails generating visual diagrams, including those without expertise, in grasping the connections among input features, model forecasts, and decision limits [22].	Particularly powerful in atmospheric and space sciences, where spatial and temporal patterns dominate (e.g., visualizing cyclone evolution, ozone depletion, or satellite sensor anomalies). Facilitates anomaly detection by making unusual outputs visually distinct. Interactive visualization allows scientists to test “what-if” scenarios by adjusting inputs and observing changes.	Poorly designed or misleading visualizations may lead to misinterpretation of the model's behaviour. Interpretation of visualizations can be subjective, and different users may derive different insights. High computational cost for rendering large-scale climate or orbital simulations.

5. CHALLENGES OF IMPLEMENTING XAI IN ATMOSPHERIC AND SPACE SCIENCES

There are some challenges underlying the fact that XAI cannot be a plug-and-play solution in atmospheric and space sciences. It must be carefully adapted, co-designed, and computationally optimized to meet the needs of scientists, policymakers, and operational users. These challenges of implementing XAI in atmospheric and space sciences are interconnected, such as data complexity, which makes it hard to extract meaningful signals; uncertainty quantification is essential to communicate forecast confidence, but remains underdeveloped in XAI, while scalability limits the operational adoption of explainability techniques, and integration of domain expertise is critical to make explanations scientifically valid and trusted. Some of the challenges and their respective solutions are presented in table 3.

Table 3: Some Challenges and Solutions of Implementing XAI in Atmospheric and Space Sciences

S/N	Challenges	Identification	Analysis	Solutions
1.	Data Complexity (DC)	Atmospheric and space science data are high-dimensional. Data often exhibit as non-stationary, changing over time due to climate variability. Large observational gaps exist e.g., sparse ground sensors in developing regions, limited in-situ measurements in space [39].	a. Traditional XAI methods like feature importance may oversimplify relationships in complex datasets. b. Dimensionality reduction techniques used for interpretability can obscure important signals, making explanations scientifically less meaningful. c. Complex correlations are difficult to represent in human-interpretable ways.	. Embed conservation laws directly into AI models to constrain explanations with physical meaning. . Transform raw sensor data into domain-relevant indices to make explanations scientifically interpretable. . Use techniques like autoencoders with physics-based constraints or causaldiscovery tools to simplify data while preserving meaningful signals. . Deploy advanced visualization adapted to climate and space datasets [40].
2.	Uncertainty Quantification (UQ)	Forecasts in weather and space sciences always involve epistemic uncertainty (from limited data and models) and aleatoric uncertainty (from inherent variability in natural systems). XAI methods typically provide explanations of	Without UQ, interpretability is incomplete; users cannot judge whether an explanation is reliable. Explaining uncertainty in a way that is accessible to decision-makers is challenging. Bayesian approaches and ensemble methods can capture uncertainty but are often computationally heavy and hard to interpret.	. Combine Bayesian neural networks with interpretability methods to provide both feature attributions and uncertainty bounds. Run multiple models, quantify spread, and apply XAI (e.g., SHAP) across ensembles for stable and uncertainty-aware insights. Extend explanation outputs beyond “why” to “how confident” (e.g.,

		what influenced a prediction but do not adequately convey *how uncertain* that prediction is [41].	Miscommunicating uncertainty can erode trust; over-confidence or under-confidence both carry risks in high-stakes applications.	confidence-weighted SHAP values). Translate uncertainty into domain-relevant formats (e.g., probability of rainfall exceeding thresholds, likelihood of solar storm intensity).
3.	Scalability	Atmospheric and space science systems run at global scales, ingesting petabytes of data daily. XAI methods like SHAP or LIME are computationally intensive and not well-suited to operational real-time forecasting pipelines.	There is a trade-off between explanation granularity and computational feasibility. Scalable explainability requires algorithms that can operate in near-real time while maintaining accuracy and depth. Visual explanations need to be generated at scale, which stresses both storage and processing resources. Lack of scalable infrastructure may confine XAI to research labs rather than operational settings.	Instead of full SHAP/LIME on every data point, use sampling or clustering to generate explanations efficiently. Train lightweight interpretable surrogates (decision trees, regression models) to mimic deep models for explanation at scale. Integrate XAI pipelines into existing HPC/operational infrastructures. Offer coarse global explanations (patterns, trends) and fine-grained local explanations selectively when critical, balancing depth and speed [42].
4.	Integration of Domain	Atmospheric and space scientists rely on physical intuition, empirical knowledge, and decades of model development. AI explanations that lack scientific grounding are often considered unhelpful.	Current XAI techniques may produce mathematically correct but scientifically irrelevant explanations. Integration of expert knowledge is needed to guide explanation design, ensuring outputs map to domain-relevant variables. Human-in-the-loop approaches are promising but resource-intensive and require co-design between AI researchers and scientists. There is a cultural challenge: scientists often expect causal explanations, while many XAI tools provide only associative insights [43].	Involve meteorologists, climate scientists, and space physicists in co-designing explanation frameworks. Develop explanation tools that highlight potential causal drivers rather than correlations alone. Domain-Specific Explanation Frameworks to outputs to scientific workflows. Interdisciplinary collaboration among AI researchers, atmospheric scientists, ethicists, and policymakers to ensure explanations serve both scientific and societal needs.

6. CONCLUSION

The survey of recent literature demonstrates that Explainable Artificial Intelligence (XAI) has emerged as a key approach for addressing the limitations of black-box artificial intelligence models in atmospheric and space sciences. While advanced machine learning and deep learning models have achieved remarkable improvements in weather forecasting, climate modeling, satellite image interpretation, air quality assessment, and space weather prediction, their lack of transparency remains a significant challenge to scientific validation, operational acceptance, and user trust. The reviewed studies consistently indicate that XAI techniques—including feature attribution methods, visualization approaches, saliency maps, surrogate models, and counterfactual explanations—enhance the interpretability of AI systems by revealing how predictions are generated and identifying the factors that most influence model decisions. These capabilities improve transparency, facilitate scientific interpretation, support model validation, and promote greater confidence among researchers, forecasters, and decision-makers.

The literature further reveals that no single XAI technique is universally applicable across all atmospheric and space science applications. Rather, the effectiveness of explainability depends on the complexity of the underlying AI model, the characteristics of the data, the operational context, and the needs of end users. Despite notable advances, several challenges remain, including balancing interpretability with predictive accuracy, explaining increasingly complex deep learning architectures, quantifying uncertainty in explanations, minimizing explanation bias, and establishing standardized evaluation metrics for explainability. These gaps highlight the need for continued research to ensure that AI systems are not only accurate but also transparent, reliable, and scientifically interpretable. Based on the reviewed literature, future research should prioritize the development of hybrid physics-informed and explainable AI models that integrate domain knowledge with advanced machine learning techniques while preserving high predictive performance. Researchers should also establish standardized evaluation frameworks, benchmark datasets, and domain-specific explainability metrics to enable consistent assessment and comparison of XAI methods. In addition, atmospheric and space science institutions should incorporate uncertainty quantification, visualization-based explanations, feature attribution techniques, and counterfactual analyses into operational AI systems to improve transparency and support evidence-based decision-making. Strengthening interdisciplinary collaboration among atmospheric scientists, space scientists, artificial intelligence researchers, software engineers, ethicists, and policymakers will be essential for developing practical standards and governance frameworks for trustworthy AI. Furthermore, greater investment in user-centered XAI tools, professional training, and decision-support systems will enhance the effective interpretation and adoption of explainable AI in operational environments.

In conclusion, the literature confirms that Explainable Artificial Intelligence serves as an essential bridge between the predictive power of black-box AI models and the scientific transparency required in atmospheric and space sciences. By embedding explainability throughout the AI development and deployment lifecycle, researchers and practitioners can improve the credibility, accountability, and operational reliability of intelligent systems. Continued advances in XAI, supported by standardized methodologies and interdisciplinary collaboration, will play a pivotal role in enabling trustworthy AI that advances scientific discovery and strengthens decision-making in weather forecasting, climate research, Earth observation, and space science applications.

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